

Dictionary Key-Value Methods

When trying to look at the information in a Python dictionary, there are multiple methods that return objects that contain the dictionary keys and values.

- `.keys()` returns the keys through a `dict_keys` object.
- `.values()` returns the values through a `dict_values` object.
- `.items()` returns both the keys and values through a `dict_items` object.

```
ex_dict = {"a": "anteater", "b":  
"bumblebee", "c": "cheetah"}  
  
ex_dict.keys()  
# dict_keys(["a", "b", "c"])  
  
ex_dict.values()  
# dict_values(["anteater", "bumblebee",  
"cheetah"])  
  
ex_dict.items()  
# dict_items([("a", "anteater"),  
("b", "bumblebee"), ("c", "cheetah")])
```

`get()` Method for Dictionary

Python provides a `.get()` method to access a dictionary value if it exists. This method takes the `key` as the first argument and an optional default value as the second argument, and it returns the value for the specified `key` if `key` is in the dictionary. If the second argument is not specified and `key` is not found then `None` is returned.

```
# without default  
{"name": "Victor"}.get("name")  
# returns "Victor"  
  
{"name": "Victor"}.get("nickname")  
# returns None  
  
# with default  
{"name": "Victor"}.get("nickname",  
"nickname is not a key")  
# returns "nickname is not a key"
```

Merging dictionaries with the `.update()` method in Python

Given two dictionaries that need to be combined, Python makes this easy with the `.update()` function. For `dict1.update(dict2)`, the key-value pairs of `dict2` will be written into the `dict1` dictionary. For keys in *both* `dict1` and `dict2`, the value in `dict1` will be overwritten by the corresponding value in `dict2`.

```
dict1 = {'color': 'blue', 'shape':  
        'circle'}  
dict2 = {'color': 'red', 'number': 42}  
  
dict1.update(dict2)  
  
# dict1 is now {'color': 'red', 'shape':  
               'circle', 'number': 42}
```

The `.pop()` Method for Dictionaries in Python

Python dictionaries can remove key-value pairs with the `.pop()` method. The method takes a key as an argument and removes it from the dictionary. At the same time, it also returns the value that it removes from the dictionary.

```
famous_museums = {'Washington':  
                  'Smithsonian Institution', 'Paris': 'Le  
Louvre', 'Athens': 'The Acropolis  
Museum'}  
famous_museums.pop('Athens')  
print(famous_museums) # {'Washington':  
                        'Smithsonian Institution', 'Paris': 'Le  
Louvre'}
```

